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High-throughput design of perpendicular magnetic anisotropy at quaternary Heusler and MgO interfaces

Sicong Jiang^{1,2} and Kesong Yang^{1,2,3}

Heusler alloys combined with MgO interfaces exhibit interfacial perpendicular magnetic anisotropy, making them attractive for energy-efficient spintronic technologies. However, finding suitable Heusler/MgO heterostructures with desired properties is challenging due to the vast range of compositions available and the complexity of interfacial structures, particularly for the emerging quaternary Heusler compounds. In this study, we report a high-throughput screening of quaternary-Heusler/MgO heterostructures for spintronic applications. By analyzing various materials descriptors, including formation energy, convex hull distance, magnetic ordering, lattice misfit, magnetic anisotropy constant, tunnel magnetoresistance, Curie temperature, and atomic site disordering, we identified 5 promising compounds out of 27,000 quaternary Heusler compounds. These compounds, namely IrCrAlTi, IrCrGaTi, IrMnZnTi, OsCrAlTa, and TaGaOsCr, show potential for designing energy-efficient perpendicular magnetic tunnel junctions. This work demonstrates an efficient approach using open quantum materials repositories, effective materials descriptors, and high-throughput computational techniques to accelerate the discovery of quaternary-Heusler-based functional materials.

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INTRODUCTION

Magnetic tunnel junctions (MTJs) have attracted increasing interest not only because of their widely existing applications in memory recording devices and magnetic sensors^{1–5} but also because of their potential applications in next-generation computing schemes such as quantum computing^{6–8}. However, traditional MTJs based on in-plane magnetic anisotropy suffer from issues of low thermal stability, high switching current, and slow magnetization switching speed. A possible solution to these issues is to use perpendicular MTJs (p-MTJs) based on perpendicular magnetic anisotropy (PMA)^{9,10}. Strength of PMA is generally measured by anisotropy energy density, i.e., magnetic anisotropy constant (K_i) in a unit of energy per unit area. As devices shrink to the nanoscale, a large positive K_i is desired to counteract the demagnetization energy and maintain sufficient thermal stability. A recent computational study suggested that when the device size scales down to 10 nm a K_i of 4.7 mJ m^{−2} is required for 10 years of data retention¹¹.

It is traditionally thought that PMA appears at material interfaces between magnetic metal layers and heavy nonmagnetic metal layers, such as Co/Pt, Co/Pd, and Co/Au, and ferromagnetic thin films such as Co, though their K_i values are generally small and less than 1.0 mJ m^{−2}^{1,2,12}. In 2010, Ikeda et al. reported a p-MTJ based on CoFeB/MgO interface with a large K_i of 1.3 mJ m^{−2}, a high tunnel magnetoresistance (TMR) ratio of 120%, and a low switching current of about 49 μ A¹³. This indicates that PMA at magnetic metals and oxides can be used to fabricate p-MTJs. Since then, to achieve desired interfacial properties for p-MTJs, numerous efforts have been made to search for alternative material interfaces consisting of different ferromagnetic materials and insulating tunnel barriers, such as Co₂FeAl/MgO^{14,15}, Mn₃Ga/MgO¹⁶, MgO/Co(111)^{17,18}, Fe/MgAl₂O₄¹⁹, Co₂FeAl/NiFe₂O₄²⁰, Fe/CuInSe₂²¹, NiFeB/MgO²².

Among all the material interfaces that have been investigated, Heusler-based heterostructures represent one class of most promising candidates for p-MTJs because of their fascinating properties^{23–29}. Heusler alloy family consists of a large number of intermetallic binary, ternary, and quaternary alloys, among which ternary full Heusler compounds are the most common, with a formula X_2YZ , where X and Y are usually transition metals and Z is typically a main-group element. Accordingly, Heusler alloys have extensive tunability through composition modifications and exhibit a wide range of properties for a variety of applications, providing a large playground for searching target functional materials including p-MTJs. In our previous work, we carried out high-throughput screening of ternary Heusler compounds in search of promising p-MTJs among hundreds of thousands of ternary Heusler compounds and successfully identified 7 candidate Heusler/MgO combinations with desired properties³⁰. Compared to the ternary Heusler compounds (full Heusler X_2YZ or half Heusler XYZ) with only three elements, quaternary Heusler compounds $XX'YZ$ with four elements have larger composition space and greater potential to find target materials^{31,32}. Moreover, the quaternary Heusler compounds generally have larger entropy than the ternary Heusler compounds, thus offering a larger potential to find more thermodynamically stable compounds. Prior high-throughput screen studies of quaternary Heusler compounds all focused on their bulk properties such as the thermoelectric and spin-gapless semiconducting properties^{33–35}. In contrast, high-throughput screen studies of material interfaces based on quaternary Heusler compounds for advanced functional applications like energy-efficient p-MTJs remain under-explored. Therefore, a systematic screening study of interfacial properties based on the quaternary Heusler alloys is highly desired to accelerate the discovery and development of functional material interfaces such as p-MTJs.

In this article, we presented a high-throughput screening approach to search for quaternary-Heusler/MgO heterostructures with desired material properties for energy-efficient p-MTJs. By

¹Department of Nano and Chemical Engineering, University of California San Diego, La Jolla, CA 92093-0448, USA. ²Program of Materials Science and Engineering, University of California San Diego, La Jolla, CA 92093-0418, USA. ³Center for Memory and Recording Research, University of California San Diego, La Jolla, CA 92093-0401, USA.

email: kesong@ucsd.edu

employing a series of effective material descriptors, we have successfully identified 5 candidate quaternary-Heusler/MgO heterostructures with robust material stability, large PMA, and high TMR ratio for p-MTJs. The atomistic origin of the large PMA in the screened heterostructures was discussed by analyzing their layer-resolved and atomic-orbital-resolved K_i values. This work paves the way for the accelerated discovery of functional materials interfaces based on the quaternary Heusler compounds through the use of open quantum materials repositories and high-throughput computational techniques.

RESULTS AND DISCUSSION

Our high-throughput screening design of the target material interfaces consists of the following three steps: (1) DATA Query, (2) Bulk Screening, and (3) Heterostructure Screening. The detailed workflow is shown in Fig. 1.

DATA query

First, we searched for thermodynamically stable quaternary Heusler compounds by querying the Open Quantum Materials Database (OQMD)³⁶. The OQMD collects formation energy (ΔE_f) and convex hull distance (ΔE_{Hull}) of the quaternary compounds, which can serve as initial screening conditions. ΔE_f of quaternary Heusler can be defined as $\Delta E_f = E_{XX'YZ} - (E_X + E_{X'} + E_Y + E_Z)$, where $E_{XX'YZ}$, E_X , $E_{X'}$, E_Y , and E_Z are total energy for compound $XX'YZ$ and for bulk ground-state X , X' , Y , and Z phase, respectively. A negative ΔE_f indicates that the compound is stable against its constituent element at 0K. ΔE_{Hull} can be described as $\Delta E_{Hull} = \Delta E_f - E_h$, where ΔE_f is formation energy and E_h is convex hull energy of the compound³⁷. ΔE_{Hull} can reflect the thermodynamic stability of a compound against its competing phases, including binary and

ternary phases. The threshold of ΔE_{Hull} can be set as high as 100 meV atom⁻¹ for quaternary Heusler compounds^{31,34}. After our DATA query of over 27,000 quaternary Heusler compounds, we selected 3094 entries based on two criteria: $\Delta E_f < 0$ and $\Delta E_{Hull} \leq 100$ meV atom⁻¹ from the OQMD database.

Bulk screening

Next, we screened bulk Heusler compounds based on other criteria, including ferromagnetic (FM) or ferrimagnetic (FI) magnetic structure, relative stability between competing phases, and lattice misfit between the quaternary Heusler compounds and MgO substrate.

Magnetic structure. The selected 3094 entries from the previous step contain both non-magnetic and magnetic compounds. To exclude the non-magnetic compounds, we set the spin magnetic moment m as one descriptor and found 776 compounds out of 3094 entries using the screening criterion $|m| > 0.2 \mu_B$ per formula unit. Since OQMD only provides data for ferromagnetic (FM) configurations, we further considered the ferrimagnetic (FI) and antiferromagnetic (AFM) configurations of these 776 compounds. Note that for the AFM and FI magnetic structures, we considered the opposite spin configurations not only between different elements but also the same element at different atomic sites. The magnetic configurations of FM, FI, and AFM of these 776 entries were calculated using high-throughput DFT calculations with the AFLOW code. The detailed magnetic structure configurations were shown in Supplementary Fig. 1. After this step, 484 compounds were confirmed with FM ground states; 80 compounds were confirmed with FI ground states; and 202 compounds were confirmed with AFM ground states, see the details in Supplementary Table 1.

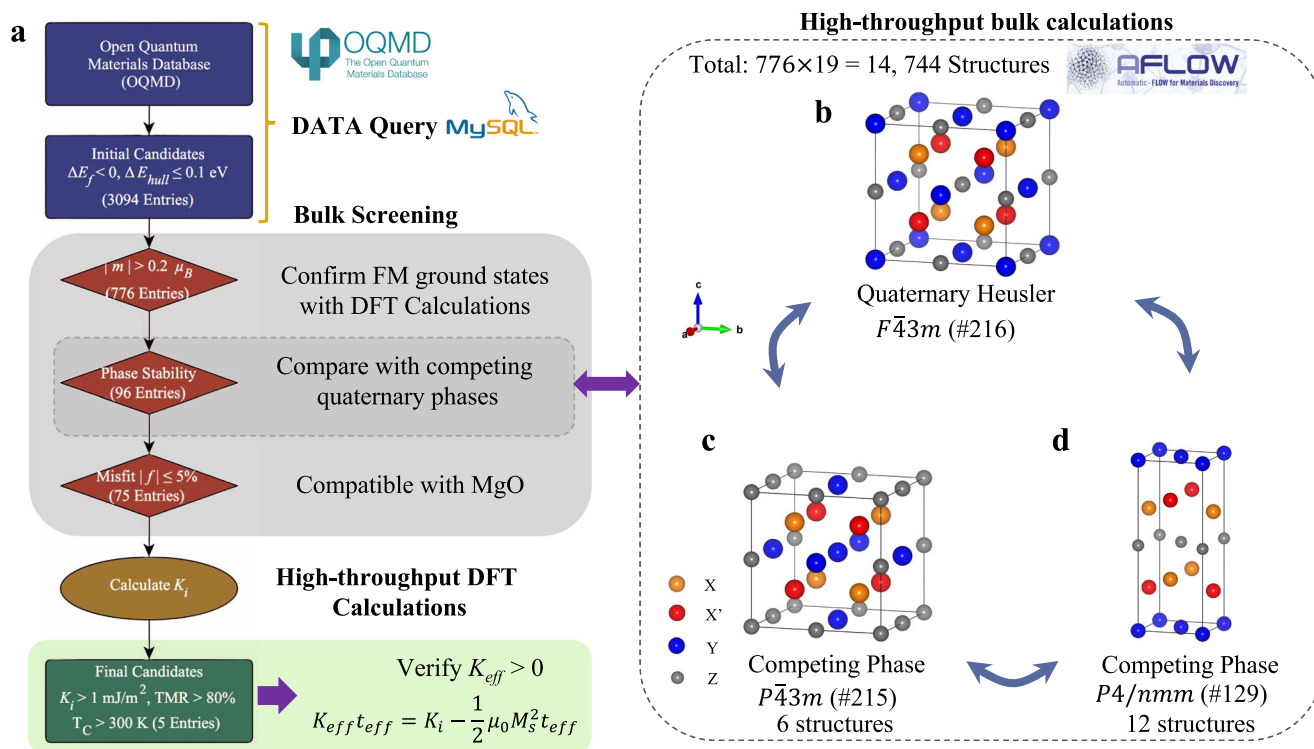


Fig. 1 Schematic workflow of high-throughput screening of bulk quaternary Heusler compounds for promising energy-efficient p-MTJs. **a** A total of 3094 entries out of 27,000 quaternary Heusler compounds were initially selected from the Open Quantum Materials Database (OQMD) using MySQL. The screening process employed materials descriptors such as formation energy (ΔE_f), convex hull distance (ΔE_{Hull}), lattice mismatch with MgO (f), interfacial cleavage energy (E_{cleav}), magnetic anisotropy constant (K_i), tunnel magnetoresistance (TMR), and Curie temperature (T_C). **b** Quaternary Heusler structure with space group of $F\bar{4}3m$ (#216). **c** Competing quaternary phase structure with space group of $P\bar{4}3m$ (#215). **d** Competing quaternary phase structure with space group of $P4/nmm$ (#129).

Phase stability. We further evaluated energetic stability for all the screened 776 quaternary Heusler compounds by comparing their total energies with those of their competing phases. As shown in Fig. 1b, quaternary Heusler has a face-centered cubic lattice with a space group of $F\bar{4}3m$ (#216), where X and X' atoms occupy the Wyckoff positions 4c (1/4, 1/4, 1/4) and 4d (3/4, 3/4, 3/4), respectively, and Y and Z atoms occupy 4a (0, 0, 0) and 4b (1/2, 1/2, 1/2), respectively. Accordingly, for a given composition of four elements (X, X', Y, and Z), the constraint of the space group symmetry leads to three unique structures³², as shown in the Supplementary Fig. 2. It is worth mentioning that relative stability between the three unique structures of quaternary Heusler compounds ($F\bar{4}3m$, #216) with the same composition has been addressed in the first step of querying data of OQMD, in which the most stable structure was selected by comparing their total energies.

In addition, it is noted that ΔE_{Hull} were calculated based on the binary, ternary, and part of quaternary compounds due to the limited data points of quaternary phases in the quantum materials repository^{36,38}. Hence, it is essential to further compare all the possible competing phases of quaternary compounds in order to select the most stable target compounds. To do this, we analyzed the structural and compositional features for all the quaternary compounds with a composition ratio of 1:1:1:1 that are both collected in the Inorganic Crystal Structure Database (ICSD) and AFLOWLIB database³⁸. After excluding the compounds containing main-group elements with high electronegativity, including O, S, N, and B as well as the light element H, we identified two competing phases of quaternary Heusler alloys with a space group of $P\bar{4}3m$ (#215) and $P4/nmm$ (#129), as shown in Fig. 1c, d, respectively.

According to symmetry analysis, $P\bar{4}3m$ (#215) has 6 unique structures while $P4/nmm$ (#129) has 12 unique structures for a given composition of four elements, which leads to a total amount of 19 structures including the one for $F\bar{4}3m$ (#216). The schematic illustration of these 19 crystal structures is shown in the Supplementary Fig. 2. Accordingly, there are a total number of 14, 744 competing phases (776×19) for the screened 776 quaternary Heusler compounds, including themselves. We then carried out high-throughput first-principles calculations for all these 14, 744 bulk structures. If the total energy of the structure is lower than its competing structures by the room-temperature thermal energy (about 25 meV atom⁻¹), then it is considered a stable structure. By using this criterion, 96 compositions were confirmed with stable quaternary Heusler structure with a space group of $F\bar{4}3m$ (#216), 21 compositions were confirmed with stable $P\bar{4}3m$ (#215) structures, 44 compositions were confirmed with stable $P4/nmm$ (#129) structures.

Lattice misfit (f). We next calculated lattice misfit between bulk quaternary Heusler and MgO substrate with the following formula: $f = (a_f - a_s)/a_s$, where a_f and a_s are lattice constants of film and substrate, respectively. The sign of f represents the strain of quaternary Heusler film undergoes from the MgO substrate, where positive and negative signs indicate compressive and tensile strain, respectively. It is generally considered that a relatively large misfit can increase the strain in the film layer and make it unstable and less likely to grow on the substrate in a high-quality crystalline structure. In this work, we selected compounds with $|f| \leq 5\%$ based on our previous empirical experience^{39,40}. After this step, 75 candidate compounds out of 96 FM or FI quaternary Heusler compounds were found suitable to build up heterostructure models with MgO.

Heterostructure screening

In this section, we studied interfacial properties of quaternary-Heusler/MgO heterostructures, focusing on their magnetic anisotropy (K_i), cleavage energy ($E_{cleav.}$), and spin polarization (P_f).

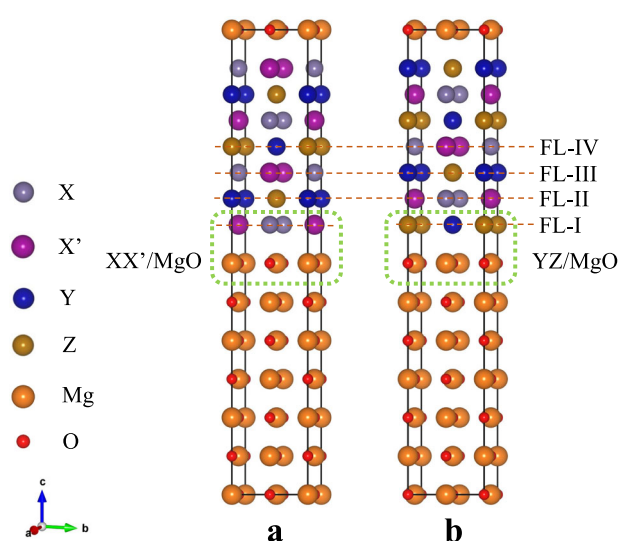


Fig. 2 Illustration of quaternary-Heusler/MgO heterostructure models. **a** XX'/MgO and **b** YZ/MgO . The MgO substrate and quaternary Heusler film were both set as seven layers. FL-I, FL-II, FL-III, and FL-IV represent the first, second, third, and fourth layers from the quaternary Heusler/MgO interface, respectively.

Magnetic anisotropy (K_i). We built up heterostructure models with seven layers of quaternary Heusler film and seven layers of MgO substrate to resemble MgO-based MTJs. This is based on prior study that seven layers of MgO substrate are enough to converge the magnetic anisotropy^{12,41}. To match lattice parameter of MgO substrate of 4.215 Å, 45° rotation was made around the [001] direction of the bulk structure of conventional quaternary Heusler. The quaternary-Heusler/MgO heterostructure has two types of symmetric interfaces, including XX'/MgO and YZ/MgO , where their schematic structures were shown in Fig. 2a, b, respectively. Note that 10 out of 75 compounds were removed from the list since they contain radioactive element Tc and not suitable for device applications. Accordingly, we calculated K_i for both XX'/MgO and YZ/MgO models of the remaining 65 bulk candidates. Table 1 shows calculated results for the 21 selected candidates, where at least one interface model has a K_i larger than 0.5 mJ m⁻². For the interface models AlTiOsMn/MgO, IrMnZnTi/MgO, OsCrGaNb/MgO, and TaGaOsCr/MgO, both terminations have PMA and at least one interface model has a K_i larger than 1 mJ m⁻². Meanwhile, in the case of IrMnZnTi/MgO and MnGaPtFe/MgO, a much large K_i of 5.00 mJ m⁻² and 3.89 mJ m⁻², were calculated for IrMn and PtGe termination, respectively, compared to the K_i of 1.2–2.1 mJ m⁻² in the Fe/MgO system^{42–44}. Interestingly, the screened compound GeMnCoFe has been successfully synthesized in a prior experimental study⁴⁵, though its interfacial K_i is rather small.

Cleavage energy ($E_{cleav.}$). Since there exist two interface models for each quaternary Heusler compound, we further analyzed their interfacial stability using cleavage energy to determine the energetically more favorable interfaces. The cleavage energy of quaternary-Heusler/MgO heterostructures can be defined as^{12,30}:

$$E_{cleav.} = (E_{slab}^{XX'YZ} + E_{slab}^{MgO} - E_{HS}^{XX'YZ/MgO}) / (2A), \quad (1)$$

where $E_{slab}^{XX'YZ}$, E_{slab}^{MgO} , and $E_{HS}^{XX'YZ/MgO}$ are the total energy of $XX'YZ$ slab, MgO slab, and $XX'YZ/\text{MgO}$ heterostructure, respectively. A is the in-plane interfacial area and the factor 2 in the denominator indicates two symmetrical interfaces. A larger $E_{cleav.}$ indicates that more energy is required to cleave the interface between quaternary Heusler film and MgO substrate. Hence, the interface is energetically more favorable and experimentally more feasible

Table 1. Summary of candidate quaternary Heusler compounds with descriptors: $\Delta E_{Hull} \leq 0.1$, $-5\% \leq f \leq 5\%$, and $K_i \geq 0.5$ for at least one termination.

Compounds	ΔE_{Hull}	f (%)	K_i		$E_{cleav.}$		P_f (P_f^{sp})		TMR (TMR^{sp})	
			XX'	YZ	XX'	YZ	XX'	YZ	XX'	YZ
AlTiOsMn [⊙]	0.020	1.4	0.58	1.76	87.19	104.41	–	0.68(0.43)	–	170(46)
GaTiLiCo	0.047	–1.6	–0.04	0.61	79.01	81.89	–	–	–	–
GeMnCoFe*	0.013	–4.3	0.16	0.71	48.81	109.99	–	–	–	–
IrCrAlTi [⊙]	0.045	2.2	1.77	0.43	102.20	81.55	0.69(0.58)	–	184(102)	–
IrCrGaTi [⊙]	0.039	2.2	1.45	0.15	100.06	66.15	0.68(0.59)	–	173(111)	–
IrCrGeLi	0.089	0.0	2.50	3.17	99.74	31.77	0.02(0.25)	–	0(13)	–
IrMnZnTi [⊙]	0.083	1.6	5.00	0.77	79.37	76.05	0.63(0.26)	–	135(15)	–
MnGaPtFe	0.078	0.2	–0.68	3.89	50.29	70.67	–	0.46(0.14)	–	52(4)
OsCoAlCr	0.062	–1.6	0.01	2.24	126.41	87.11	–	–	–	–
OsCrAlNb	0.051	3.8	1.84	0.41	115.84	68.82	0.51(0.22)	–	71(10)	–
OsCrAlTa [⊙]	0.058	3.7	1.68	–0.53	115.95	74.18	0.63(0.42)	–	130(43)	–
OsCrAlTi	0.019	2.6	0.02	1.16	117.21	86.73	–	–	–	–
OsCrAlV [⊙]	0.054	0.6	1.68	–0.08	118.53	96.24	0.58(0.46)	–	101(53)	–
OsCrGaNb	0.071	3.9	2.05	0.96	114.21	57.80	0.53(0.3)	–	79(19)	–
OsCrGaTi	0.088	2.7	0.37	0.88	117.58	67.92	–	–	–	–
OsCrZnNb	0.090	4.1	–0.02	0.87	116.37	69.92	–	–	–	–
OsCrZnTa	0.083	4.0	–0.65	0.60	116.26	75.69	–	–	–	–
RuMnGaSc	0.079	4.6	0.63	0.83	86.78	46.69	–	–	–	–
TaGaOsCr [⊙]	0.085	3.8	0.73	1.40	64.50	114.07	–	0.62(0.44)	–	125(48)
TiZnIrCr	0.049	2.5	0.95	–0.46	76.91	104.35	–	–	–	–
VGaOsCr [⊙]	0.077	0.8	–0.28	1.88	73.92	118.64	–	0.51(0.38)	–	69(32)

⊙ indicates bulk compounds with FI ground state, ⊙ represents the target quaternary Heusler in the final list, and * shows the experimentally synthesized compounds.

ΔE_{Hull} and f refer to the convex hull distance (eV atom^{–1}) and the lattice misfit with MgO substrate, respectively. K_i , $E_{cleav.}$, P_f (P_f^{sp}), and TMR (TMR^{sp}) indicate magnetic anisotropy constant (mJ m^{–2}), cleavage energy (meV Å^{–2}), spin polarization (spin polarization only from s and p orbital), and tunnel magnetoresistance ratio (tunnel magnetoresistance ratio only from s and p orbital) (%), respectively. P_f was calculated only for the candidate quaternary-Heusler/MgO heterostructures whose energetically more favorable interface has a $K_i > 1$ mJ m^{–2}.

than those with a smaller $E_{cleav.}$. If two interfaces have close $E_{cleav.}$, both interfaces are considered as experimentally feasible. The interface models with PMA for compounds OsCoAlCr, OsCrAlTi, OsCrZnNb, OsCrZnTa, and TiZnIrCr were found to have a lower $E_{cleav.}$ than those without PMA, and thus these interfaces were considered as energetically not favorable and excluded from our final list.

Spin polarization (P_f). We next estimated spin polarization (P_f) for heterostructures with $K_i > 1$ mJ m^{–2} in the stable interface models using the following equation⁴⁶:

$$P_f = \frac{D_+(E_F) - D_-(E_F)}{D_+(E_F) + D_-(E_F)}, \quad (2)$$

where $D_+(E_F)$ and $D_-(E_F)$ are the density of states for spin-majority and spin-minority channels at Fermi level, respectively. TMR ratio was then calculated with the Julliere's equation⁴⁷:

$$TMR = \frac{2P_1P_2}{1 - P_1P_2}, \quad (3)$$

where P_1 and P_2 are spin polarization of the two magnetic electrodes. For Heusler/MgO-based MTJs, we assumed that the two electrodes consist of the same quaternary Heusler compound, thus leading to $P_1 = P_2 = P_f$. Here, we calculated P_f only for the candidate quaternary-Heusler/MgO heterostructures whose energetically more favorable interface has a $K_i > 1$ mJ m^{–2}, as listed in Table 1. After calculating P_f and TMR ratio, the compounds IrCrGeLi, MnGaPtFe, OsCrAlNb, OsCrGaNb, and VGaOsCr were removed from our final list due to the low spin polarization and

low TMR < 80%. In particular, in the case of IrCrGeLi/MgO, though the K_i of the IrCr/MgO interface is 2.50 mJ m^{–2}, the spin polarization and estimated TMR is very small. This could be related to the presence of Li atoms since previous theoretical calculations have shown that half Heusler compounds LiMnZ ($Z = N, P, Si$) containing Li atoms are ordinary ferromagnets without half-metallicity.⁴⁸ Previous studies on the off-stoichiometric Heusler alloys Co₂MnSi and Co₂(Mn,Fe)Si have shown that the TMR ratio can be accurately described by the sp spin polarization, which only considers s and p orbitals, instead of the total spin polarization that includes s , p , and d orbitals.^{49,50} Consequently, in this study, we also calculated the sp polarization (P_f^{sp}) and the resulting TMR ratio (TMR^{sp}) for comparison purposes.

Effective magnetic anisotropy (K_{eff}). Next we calculated effective magnetic anisotropy (K_{eff}) to evaluate the robustness of the interfacial PMA by considering shape anisotropy (K_{shape}). This is because shape anisotropy spontaneously favors in-plane spin alignment for the finite thin films. The K_{eff} can be defined as¹:

$$K_{eff} = \frac{K_i}{t_{eff}} - \frac{1}{2}\mu_0 M_s^2 \quad (4)$$

where $\frac{1}{2}\mu_0 M_s^2$ represents the shape anisotropy per unit volume, t_{eff} is the thickness of the FM or FI layer, μ_0 is the magnetic constant, and M_s is the saturation magnetization per unit volume. At this point, 7 candidate systems, including AlTiOsMn, IrCrAlTi, IrCrGaTi, IrMnZnTi, OsCrAlTa, OsCrAlV, and TaGaOsCr, are selected. Our calculations confirmed that all these screened Heusler/MgO heterostructures have a positive K_{eff} , suggesting a robust PMA.

It is noted that some quaternary Heusler compounds have been theoretically predicted or synthesized in prior experiments but they were excluded from our final list because they cannot form the desired interfacial properties. For example, previously theoretically predicted compound FeCrTiAl^{35} and experimentally reported compound CoRhMnGe^{51} were screened in our list of 75 compounds, but not in our final list due to their in-plane magnetic anisotropy in the MgO-based heterostructures. Another quaternary Heusler, CoFeMnSi , synthesized in previous experiment, is not included in our list because of the large lattice misfit with MgO substrate⁴⁵. Besides, our calculations for the composition CoFeMnSi show that the energy difference between the space group #216 phase and #215 phase is less than 15 meV atom^{-1} . Thus, it is expected that the quaternary regular Heusler phase of CoFeMnSi may not be stable, which is also consistent with the reported D0_3 disordering in the literature^{45,52}. A prior experimental study shows that CoFeCrGe crystallizes in a fully ordered Y-type structure (regular Heusler structure with space group $F\bar{4}3m$, #216)⁵³. Interestingly, our calculations show that its competing phase with the space group #215 is more stable because of lower total energy. This also correlates with previous experimental findings that the compound CoFeCrGe exhibits a small amount of secondary phase after annealing⁵³ or becomes other compounds⁵⁴, thus worthy of further experimental and theoretical studies. A recent computational study proposes that quaternary Heusler alloys IrCrMnAl and IrCrMnGa are promising for building MgO-based MTJs due to their high Curie temperatures⁵⁵. However, our calculations show that these two compounds are energetically unfavorable compared to their competing phases with space group no. #215, and thus not selected in our final list.

Curie temperature. Next, we estimated Curie temperature T_C for the screened seven candidate compounds based on the mean-field approximation using the equation $k_B T_C = 2/3 \Delta E$ ^{56,57} where k_B is the Boltzmann constant and ΔE is the total energy difference between the ferromagnetic and antiferromagnetic spin configurations. This approach maps the DFT energetics onto the classical Heisenberg model $H = \sum_{ij} J_{ij} \vec{m}_i \cdot \vec{m}_j$, where J_{ij} represents the magnetic exchange interaction between atomic sites i and j possessing local magnetic moments $\vec{m}_i$ and $\vec{m}_j$, respectively. Our findings indicate that two of the screened candidate compounds, AlTiOsMn and OsCrAlV , have Curie temperatures (T_C) of 277 K and 264 K, respectively, and are therefore excluded from our final list. The remaining five compounds exhibit T_C values well above room temperature. Table 2 presents a summary of these five compounds in our final list.

Disordering effects. We next evaluated possible disordering effects in the 5 selected final candidates by calculating the exchange energy between any two swapping atoms. Since the quaternary Heusler compound has four types of atoms, there are six combinations between atom swapping. Here we took the case of IrCrAlTi as one example to discuss the computational procedure and analysis. In the case of IrCrAlTi , there are 6 possible combinations to swap any 2 atoms, including Al-Cr, Al-Ir, Cr-Ir, Ti-Al, Ti-Cr, and Ti-Ir. The calculated total energy as a function of the percentage of the disorder degree for all the selected 5 final quaternary Heusler compounds are shown in the Supplementary Figs. 3 and 4. In the swapping case of Al-Cr, 50% disorder means swapping 2 Al atoms with 2 Cr atoms, and 100% disorder means swapping 4 Al atoms with 4 Cr atoms. Accordingly, 0% and 100% disorder correspond to ordered structures. Note that the calculated function curve is symmetric along the boundary line of the 50% disorder for the Cr-Ir and Ti-Al swapping combinations because of the symmetry of space group of $F\bar{4}3m$ (#216). If the total energy of disordered structures is lower than the ordered structures by larger than 25 meV atom^{-1} , then the compounds tend to be disordered; otherwise, the compounds tend to be

Table 2. Summary of 5 final candidate Heusler/MgO heterostructures with materials parameters: K_i (mJ m^{-2}), saturation magnetization M_s (emu cm^{-3}), shape anisotropy K_{shape} ($\times 10^7 \text{ erg cm}^{-3}$), effective anisotropy per unit volume K_{eff} ($\times 10^7 \text{ erg cm}^{-3}$), and Curie temperature T_C (K).

Compounds	Interface	K_i	M_s	K_{shape}	K_{eff}	T_C
IrCrAlTi	IrCr	1.77	493	0.15	1.59	505
IrCrGaTi	IrCr	1.45	495	0.15	1.27	604
IrMnZnTi	IrMn	5.00	643	0.26	4.48	754
OsCrAlTa	OsCr	1.68	496	0.15	1.45	338
TaGaOsCr	OsCr	1.40	545	0.19	1.26	342

ordered. By using this approach, we confirmed that all the selected 5 quaternary Heusler compounds are in favor of ordered structure. In the case of IrMnZnTi , the Ti-Zn disordered structure has the lowest total energy, but it was still considered as an ordered structure because the energy difference between the ordered and disordered structures is less than 25 meV atom^{-1} .

Origin of PMA

Lastly, we investigated the atomistic origin of PMA at quaternary-Heusler/MgO interfaces for the screened 5 final candidate systems. Interestingly, a careful examination shows that there are four common magnetic ions including 3d Mn and Cr and 5d Os and Ir at all the considered interfaces, see Table 2. To reveal the role of these magnetic ions in producing the interfacial PMA, we calculated layer-resolved K_i for these screened Heusler/MgO heterostructures, see Fig. 3. Note that IrCrGaTi/MgO exhibits similar layer-resolved K_i with IrCrAlTi/MgO and thus the results of the former system are not shown here. Our calculations show that the PMA mainly comes from the interfacial 5d Ir and Os atoms. In the case of IrCrAlTi/MgO and IrMnZnTi/MgO , the PMA results from the Ir atoms in the FL-I and FL-III layers, see Fig. 3a, b, respectively. In the case of OsCrAlTa/MgO and TaGaOsCr/MgO , the interfacial Os atoms in the FL-I layer mainly contribute to the PMA, as shown in the Fig. 3c, d.

To further understand the microscopic origin of the interfacial PMA, we calculated the orbital-resolved K_i for the interfacial atoms, see Fig. 4. Figure 4a, b shows orbital hybridization of Ir 5d orbitals in IrCrAlTi/MgO and IrMnZnTi/MgO models, respectively. The PMA is found to mainly come from the hybridization between d_{yz} and $d_{x^2-y^2}$, and d_{yz} and d_{z^2} states. In the case of OsCrAlTa/MgO and TaGaOsCr/MgO , the orbital hybridization between $d_{x^2-y^2}$ and d_{xy} , and $d_{x^2-y^2}$ and d_{yz} states of the interfacial Os atom leads to the out-of-plane magnetic anisotropy, while the orbital hybridization between d_{z^2} and d_{yz} states produces the in-plane magnetic anisotropy, see Fig. 4c, d, respectively.

METHODS

The automatic framework AFLOW⁵⁸ based on Vienna Ab-initio Simulation Package (VASP)⁵⁹ was used for high-throughput density functional theory (DFT) calculations of bulk quaternary compounds. The projector-augmented wave (PAW) pseudopotentials were used for electron-ion interactions⁶⁰. The generalized gradient approximation (GGA) parameterized by Perdew-Burke-Ernzerhof (PBE) was used for exchange-correction functional⁶¹. A convergence threshold of $0.01 \text{ meV atom}^{-1}$ and a k -points grid with a separation of $0.05 \text{ \AA}^{-1}$ were set for structural relaxation. A denser k -points grid with a separation of $0.04 \text{ \AA}^{-1}$ was used for static total energy calculations. Other computational settings such as cutoff energy were automatically generated by the AFLOW code for both structural relaxation and static calculations⁵⁸. In the heterostructure calculations, Γ -centered k -points grids were set to $8 \times 8 \times 1$ and $13 \times 13 \times 1$ for structural relaxation and static

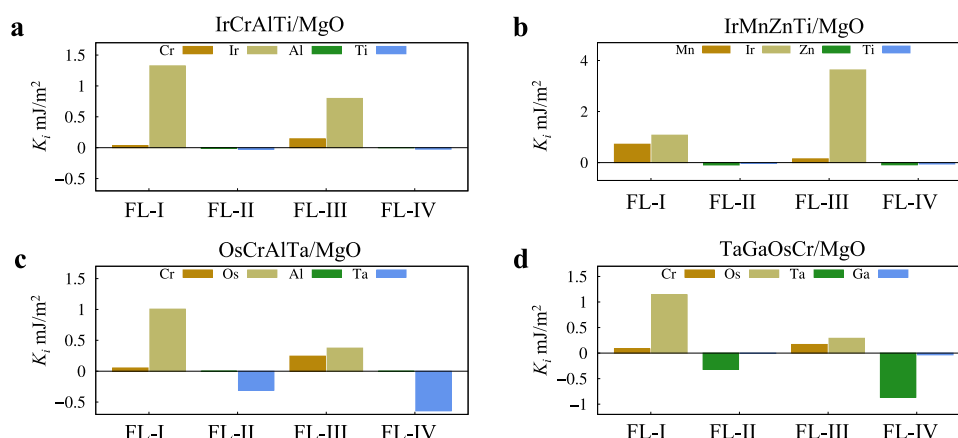


Fig. 3 Calculated layer-resolved K_i for selected quaternary-Heusler/MgO models. **(a)** IrCrAlTi/MgO with Ir as the interface, **(b)** IrMnZnTi/MgO with Ir as the interface, **(c)** OsCrAlTa/MgO with Os as the interface, and **(d)** TaGaOsCr/MgO with Os as the interface. FL-I, FL-II, FL-III, and FL-IV correspond to the first, second, third, and fourth layers of quaternary Heusler film from the interface.

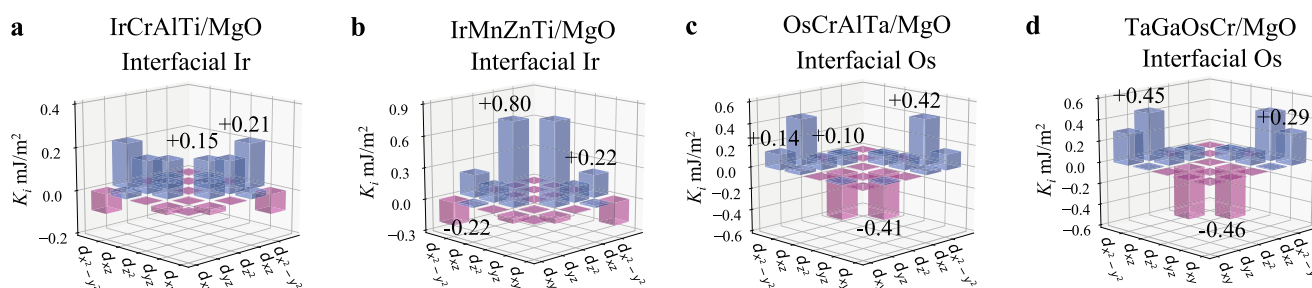


Fig. 4 Calculated orbital-resolved K_i for selected quaternary-Heusler/MgO models. **(a)** IrCrAlTi/MgO, **(b)** IrMnZnTi/MgO, **(c)** OsCrAlTa/MgO, and **(d)** TaGaOsCr/MgO.

calculation, respectively. All the lattice parameters and atomic positions were fully relaxed until the residual forces were smaller than the threshold of $5 \text{ meV } \text{\AA}^{-1}$. The self-consistent field calculations were performed with a convergence threshold of 10^{-6} eV . The magnetic anisotropy constant K_i was calculated as $(E_{[100]} - E_{[001]})/(2A)^{12}$, where $E_{[100]}$ ($E_{[001]}$) represents the total energy including SOC with magnetization along $[100]$ ($[001]$) direction. The factor of 2 in the denominator represents two symmetrical interfaces in the heterostructure model and A is the in-plane area.

DATA AVAILABILITY

The data that support the findings of this study are available in the [Supplementary information](#) document. The structural files of the selected 776 thermodynamically stable quaternary Heusler compounds are provided online at <https://github.com/ksyang2013/Heusler>. Other relevant data are available from the corresponding author upon reasonable request.

CODE AVAILABILITY

The code developed in this study is available for access in the GitHub repository, <https://github.com/ksyang2013/Heusler>.

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AUTHOR CONTRIBUTIONS

S.J. collected the materials data from the open quantum materials database, performed high-throughput calculations and screening, and developed an automated framework to evaluate disorder effects. K.Y. designed and supervised the research project. All authors contributed to the manuscript writing.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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Correspondence and requests for materials should be addressed to Kesong Yang.

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